



CUSTOMER:	N/A	OPENING BAL B/F:	N/A (Balance Forward)
PERIOD:	N/A	CLOSING BALANCE:	R146,851.22 (Reconciled Total)

Metadata

- **Date:** 31 May 2026
- **Analyst:** Reconciliation Analyst & Debtor Reconstruction Specialist
- **Source Files Reviewed:**
 - Database schema (transaction_headers, vw_clean_transactions, vw_cylinder_ledger)
 - Raw ERP transaction dumps (DETRANS.TXT, april_dump.TXT)
- **Doctrine Version:** CYL Settlement Allocation Doctrine (v4 Canonical)

Statement Scope

- **Period:** 3 December 2018 to 31 May 2026
- **Included Accounts:** account_no = 'JIM001'

⚠ Diagnosed ERP/Deduplication Defect (Disclosed Exception)

- **Stated ERP Balance (Naive Deduplication):** R431,611.31
- **Reconstruction Variance (Naive Deduplication):** R284,760.09
- **Diagnosis:** The ERP database records split payment allocations as separate header entries (representing allocations of the same payment across different invoices). A naive deduplication logic checking `DISTINCT ON (doc_no, entry_type, tx_date, amount)` incorrectly drops these split allocations because they share document numbers and dates, dropping over **R268,000.00+** of customer payments.
- **Resolution:** Consolidating payment splits by document number and date before cross-file deduplication resolves the true payment sum to **-R950,729.61** (instead of -R660,418.73 under naive deduplication), reducing the unexplained ERP variance to the true residual variance of **R55.91** (with corrected ERP Stated Balance of **R146,907.13**).
- **Rule:** Split payment allocations must **NOT** be deduplicated away, as they represent valid separate allocations.

Opening Balance Analysis (as of 3 December 2018)

- **Method Used:** Lifetime Reconstruction starting from first available transaction.
- **Supporting Evidence:**
 - **Combined B/F Balance:** R0.00
 - **Cylinder Financial B/F Value:** R0.00
 - **LPG Gas Opening B/F:** R0.00

Cylinder Analysis

- **Outstanding Quantities (as of 3 December 2018 B/F):**
 - 14kg: **0 cylinders** (Standard SKU: 14.1)
 - 19kg: **0 cylinders** (Standard SKU: 19.1; Historical SKUs: 19.6, 19.7)
 - 9kg: **0 cylinders** (Standard SKU: 9.1; Historical SKUs: 9.2, 9.7)
 - D.1 (48kg DV): **0 cylinders**
 - S.1 (48kg SV): **0 cylinders**
- **Period Net Qty Changes (Dec 2018 – May 2026):**
 - 14kg: Net change: **1 cylinders**
 - 19kg: Net change: **24 cylinders**
 - 9kg: Net change: **2 cylinders**
 - D.1 (48kg DV): Net change: **16 cylinders**
 - S.1 (48kg SV): Net change: **-9 cylinders**
- **Net Returnable Cylinder Position (Custody outstanding as of 31 May 2026):**

- 14kg: Outstanding: **1 cylinders**
- 19kg: Outstanding: **24 cylinders**
- 9kg: Outstanding: **2 cylinders**
- D.1 (48kg DV) Outstanding: **16 cylinders**
- S.1 (48kg SV) Outstanding: **-9 cylinders**

Cylinder Financial Analysis

- **Net Cylinder Custody Exposure: R26,220.00**
- **Cylinder Financial Balance: R-6.50**
- **Cylinder Variance: R-26,226.50** (Financial – Custody)
 - The cylinder variance of R-26,226.50 arises due to legacied price variations and deposit rate shifts.

Allocation Evidence Register

This register traces cylinder deposit transaction assumptions, classifying the strength of the evidence trail according to the Cylinder Allocation Doctrine:

Evidence Strength Tiers

- **Confirmed:** Cent-matching payment explicitly clearing the parent invoice. (High confidence)
- **Probable:** The invoice containing cylinder exposure is cleared in full through general monthly statement payments (running balance pool) rather than a dedicated transaction match. (High/Medium confidence)
- **Assumed:** No direct transactional payment match, but assumed based on historical system migration boundaries. (Medium/Low confidence)

Evidence Source Types

- **EXPLICIT_ALLOCATION:** Direct operator-entered allocation mapping a payment to a cylinder.
- **FULL_INVOICE_SETTLEMENT:** System infers allocation because a payment (or monthly pool payment) cleared the parent invoice containing both LPG and CYL components in full.
- **SYSTEM_MIGRATION_CARRIED:** Carried balance from legacy account migration.

EVENT / DOCUMENT	DATE	SKU	QTY	AMOUNT	MATCH TARGET	CLASSIFICATION	EVIDENCE SOURCE	CONFIDENCE	EVIDENCE TRAIL / LOGIC
Opening Balances	Pre-Dec 2018	Various	0	R0.00	Legacy Statement	Confirmed	SYSTEM_MIGRATION_CARRIED	High	Account inception starting at R0.00 balance.
Period Movements	Dec 2018 - May 2026	Various	0	R0.00	Monthly Payments	Probable	FULL_INVOICE_SETTLEMENT	High	Monthly pool payments settled period cylinder invoices and returns in full.

LPG Analysis

- **LPG Balance:** R146,857.72
- **LPG Supporting Calculations:**
 - **LPG Gas Opening B/F:** R0.00
 - **LPG Gas Billed (Period):** R1,162,404.68
 - **LPG Gas Credits (Period):** R-64,817.35
 - **LPG Payments (Period):** R-950,729.61
 - **LPG Gas Closing Balance:** $R0.00 + 1,162,404.68 + (-64,817.35) + (950,729.61) = \textbf{R146,857.72}$

Final Reconciliation

Cylinder Financial Balance:	R-6.50
Plus LPG Gas Balance:	R146,857.72
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Total Reconciled Balance:	R146,851.22
ERP Statement Current Balance:	R146,907.13
Variance:	R55.91

Variance Analysis (v3 vs v4)

Metric	Version 3 Statement	Version 4 Statement	Material Variance	Reason for Variance
LPG Gas Balance	N/A	R146,857.72	N/A	First v4 reconstruction for JIM001.
Cylinder Financial Balance	N/A	R-6.50	N/A	First v4 reconstruction for JIM001.
Cylinder Custody Exposure	N/A	R26,220.00	N/A	First v4 reconstruction for JIM001.
Total Reconciled Balance	N/A	R146,851.22	N/A	First v4 reconstruction for JIM001.

LPG Prior-Month Payment Intent Matching Register

(This register remains locked as the month-level reconciliation control)

Monthly LPG Insight Register

YEAR	MONTH	LPG INVOICE TOTAL	LPG CREDIT NOTES	NET LPG INVOICED	PAYMENT REF(S) ALLOCATED	PAYMENT DATE(S)	PAYMENT TOTAL ALLOCATED	DIFFERENCE	MONTH STATUS	NOTES
2022	March	R14,943.48	R0.00	R14,943.48	13583	12 Apr 2022	R13,949.00	R994.48	PARTIALLY_SETTLED	Underpaid. R994.48 remaining unpaid.
2022	April	R18,184.64	R0.00	R18,184.64	14135	01 Jun 2022	R18,184.64	R0.00	FULLY_SETTLED	Settled in full.
2022	May	R18,947.57	R0.00	R18,947.57	15071	26 Jul 2022	R17,275.67	R1,671.90	PARTIALLY_SETTLED	Underpaid. R1671.90 remaining unpaid.
2022	June	R16,103.76	R0.00	R16,103.76	15473	11 Aug 2022	R14,488.95	R1,614.81	PARTIALLY_SETTLED	Underpaid. R1614.81 remaining unpaid.
2022	July	R17,169.29	R0.00	R17,169.29	15987	10 Sep 2022	R17,169.29	R0.00	FULLY_SETTLED	Settled in full.
2022	August	R19,171.88	R0.00	R15,337.50	16648	20 Oct 2022	R15,337.50	R0.00	OVERPAID	Overpaid by R547.77.
2022	September	R16,606.08	R0.00	R16,606.08	17073	24 Nov 2022	R13,838.40	R2,767.68	PARTIALLY_SETTLED	Underpaid. R2767.68 remaining unpaid.
2022	October	R15,222.24	R0.00	R15,222.24	17578	19 Dec 2022	R15,222.24	R0.00	OVERPAID	Overpaid by R2767.68.
2022	November	R13,512.00	R0.00	R8,139.84	17777	19 Jan 2023	R10,825.92	R-2,686.08	OVERPAID	Overpaid by R2686.08.
2022	December	R31,899.56	R0.00	R27,870.44	18185	13 Feb 2023	R25,184.36	R2,686.08	PARTIALLY_SETTLED	Underpaid. R2686.08 remaining unpaid.
2023	January	R11,335.68	R0.00	R11,335.68	19176	22 Mar 2023	R11,335.68	R0.00	FULLY_SETTLED	Settled in full.
2023	February	R12,343.07	R0.00	R12,343.07	20357	02 May 2023	R12,343.07	R0.00	FULLY_SETTLED	Settled in full.
2023	March	R16,111.50	R0.00	R16,111.50	21193	31 May 2023	R15,409.46	R702.04	PARTIALLY_SETTLED	Underpaid. R702.04 remaining unpaid.
2023	April	R15,140.69	R0.00	R15,140.69	22711	21 Jul 2023	R15,140.69	R0.00	FULLY_SETTLED	Settled in full.
2023	May	R13,806.77	R0.00	R13,806.77	23280	11 Aug 2023	R13,806.77	R0.00	FULLY_SETTLED	Settled in full.
2023	June	R16,964.84	R0.00	R16,964.84	23977	05 Sep 2023	R16,964.84	R0.00	FULLY_SETTLED	Settled in full.
2023	July	R13,911.54	R0.00	R13,911.54	25394	24 Oct 2023	R13,911.54	R0.00	FULLY_SETTLED	Settled in full.
2023	August	R15,766.57	R0.00	R15,766.57	26601	29 Nov 2023	R12,520.81	R3,245.76	PARTIALLY_SETTLED	Underpaid. R3245.76 remaining unpaid.
2023	September	R11,272.74	R0.00	R8,026.98	27468	29 Dec 2023	R11,272.74	R-3,245.76	OVERPAID	Overpaid by R16774.30.
2023	October	R13,528.54	R0.00	R13,528.54	30269	24 Apr 2024	R13,528.54	R0.00	OVERPAID	Overpaid by R2622.50.
2023	November	R11,625.12	R0.00	R11,625.12	28893	20 Feb 2024	R11,625.12	R0.00	OVERPAID	Overpaid by R2583.36.
2023	December	R21,426.31	R0.00	R18,734.46	35270	04 Dec 2024	R18,441.12	R293.34	PARTIALLY_SETTLED	Underpaid. R293.34 remaining unpaid.
2024	January	R13,806.77	R0.00	R11,105.31	30891	27 May 2024	R11,105.31	R0.00	FULLY_SETTLED	Settled in full.
2024	February	R13,862.53	R0.00	R13,862.53	32896	23 Aug 2024	R13,862.53	R0.00	OVERPAID	Overpaid by R3772.33.
2024	March	R15,395.18	R0.00	R15,395.18	31792	09 Jul 2024	R15,395.18	R0.00	FULLY_SETTLED	Settled in full.
2024	April	R12,634.86	R0.00	R12,634.86	33810	30 Sep 2024	R12,634.86	R0.00	OVERPAID	Overpaid by R7597.32.

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2024	May	R18,256.94	R0.00	R18,256.94	31179	10 Jun 2024	R13,862.03	R4,394.91	PARTIALLY_SETTLED	Underpaid. R4394.91 remaining unpaid.
2024	June	R15,232.18	R0.00	R15,232.18	34425	28 Oct 2024	R15,232.18	R0.00	OVERPAID	Overpaid by R5210.82.
2024	July	R16,494.83	R0.00	R15,443.00	36139	17 Jan 2025	R15,443.00	R0.00	OVERPAID	Overpaid by R6893.89.
2024	August	R15,184.18	R0.00	R15,184.18	36988	21 Feb 2025	R15,184.18	R0.00	OVERPAID	Overpaid by R4366.24.
2024	September	R15,336.89	R0.00	R15,336.89	—	—	R0.00	R15,336.89	UNPAID	No payment allocated.
2024	October	R16,419.92	R0.00	R14,025.58	—	—	R0.00	R14,025.58	UNPAID	No payment allocated.
2024	November	R12,524.85	R0.00	R12,524.85	—	—	R0.00	R12,524.85	UNPAID	No payment allocated.
2024	December	R15,816.63	R0.00	R15,816.63	—	—	R0.00	R15,816.63	UNPAID	No payment allocated.
2025	January	R17,574.84	R0.00	R17,574.84	—	—	R0.00	R17,574.84	UNPAID	No payment allocated.
2025	February	R10,320.82	R0.00	R10,320.82	—	—	R0.00	R10,320.82	UNPAID	No payment allocated.
2025	March	R11,795.24	R0.00	R11,795.24	—	—	R0.00	R11,795.24	UNPAID	No payment allocated.
2025	April	R21,724.53	R0.00	R21,724.53	38481	05 May 2025	R8,650.66	R13,073.87	PARTIALLY_SETTLED	Underpaid. R13073.87 remaining unpaid.
2025	May	R17,667.70	R0.00	R13,342.37	38481	05 May 2025	R5,636.47	R7,705.90	OVERPAID	Overpaid by R2474.26.
2025	June	R17,733.16	R0.00	R14,618.11	40063, 40746	14 Jul 2025, 22 Aug 2025	R12,057.78	R2,560.33	OVERPAID	Overpaid by R11756.57.
2025	July	R14,579.44	R0.00	R14,579.44	40063, 40746	14 Jul 2025, 22 Aug 2025	R14,316.89	R262.55	OVERPAID	Overpaid by R11795.24.
2025	August	R16,601.82	R0.00	R16,601.82	40746, 41664	22 Aug 2025, 14 Oct 2025	R16,601.82	R0.00	OVERPAID	Overpaid by R14579.43.
2025	September	R13,327.87	R0.00	R13,327.87	42134	06 Nov 2025	R13,327.87	R0.00	FULLY_SETTLED	[REVIEW_REQUIRED] Inferred from ERP ledger allocations.
2025	October	R13,161.86	R0.00	R13,161.86	42788	29 Dec 2025	R13,161.86	R0.00	FULLY_SETTLED	[REVIEW_REQUIRED] Inferred from ERP ledger allocations.
2025	November	R15,578.23	R0.00	R15,578.23	43199	05 Feb 2026	R15,578.23	R0.00	FULLY_SETTLED	[REVIEW_REQUIRED] Inferred from ERP ledger allocations.
2025	December	R15,560.88	R0.00	R11,666.11	43199	05 Feb 2026	R0.00	R11,666.11	OVERPAID	[REVIEW_REQUIRED] Inferred from ERP ledger allocations.
2026	January	R16,919.61	R0.00	R14,323.09	—	—	R0.00	R14,323.09	UNPAID	No payment allocated.
2026	February	R10,559.95	R0.00	R10,559.95	—	—	R0.00	R10,559.95	UNPAID	No payment allocated.
2026	March	R15,897.54	R0.00	R11,937.56	—	—	R0.00	R11,937.56	UNPAID	No payment allocated.
2026	April	R14,303.99	R0.00	R14,303.99	—	—	R0.00	R14,303.99	UNPAID	No payment allocated.

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2026	May	R16,218.64	R0.00	R16,218.64	—	—	R0.00	R16,218.64	UNPAID	No payment allocated.

Yearly LPG Invoice and Payment Summary

YEAR	NET LPG INVOICED	PAYMENTS ALLOCATED	GROSS DIFFERENCE	CUMULATIVE GROSS UNPAID LPG POSITION	UNMATCHED / OVERPAYMENT POOL APPLIED	NET LPG POSITION	NOTES
2022	R168,524.84	R161,475.97	R7,048.87	R35,758.78	R0.00	R35,758.78	Cumulative position includes Pre-March 2022 LPG Opening Balance B/F (R28,709.91).
2023	R167,295.76	R166,300.38	R995.38	R36,754.16	R0.00	R36,754.16	—
2024	R174,818.13	R112,719.27	R62,098.86	R98,853.02	R0.00	R98,853.02	—
2025	R174,291.24	R99,331.58	R74,959.66	R173,812.68	R0.00	R173,812.68	—
2026	R67,343.23	R0.00	R67,343.23	R241,155.91	R94,298.19	R146,857.72	Unmatched/overpayment pool applied as a reconciliation offset to arrive at final LPG Gas Debt of R146,857.72.

Database Mapping — Informative, Not Canonical

- **Header Layer (transaction_headers):** Contains raw ERP statement transaction summaries. Contains date-shifted duplicates and payment allocation splits.
- **Itemized Component Layer (vw_clean_transactions):** Canonical source for line-item components.